Data Analytics (DA)

Power BI Assignment 1 – Data Transformation & Data Modeling

Overview:

Business Problem:

As a Data Analyst, you will continue working with the E-Commerce Sales Dataset used in the previous assignment. The dataset already contains the required tables and relationships created during data transformation and data modeling. In this assignment, you will use DAX (Data Analysis Expressions) in Power BI to create calculated columns and measures that generate meaningful business metrics. You will then develop interactive visualizations to analyze sales trends, profitability, and performance indicators within the e-commerce business environment. The focus of this assignment is on DAX calculations, analytical thinking, and visual storytelling, while ensuring that the underlying data model supports accurate reporting and insights.

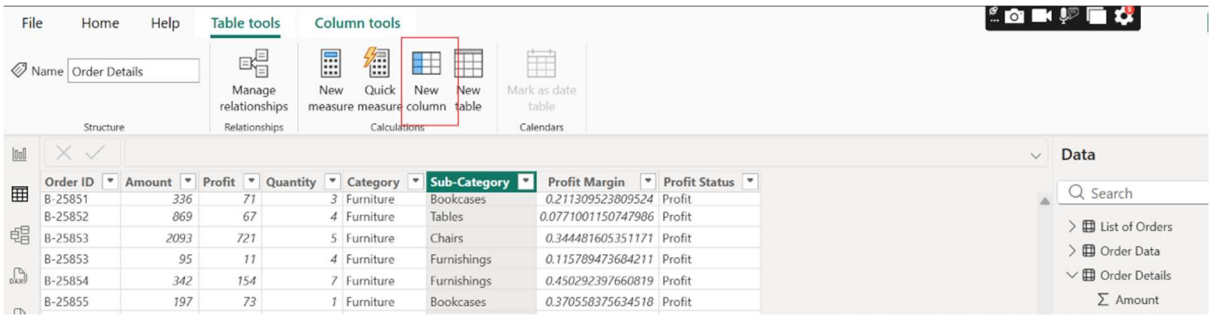
Dataset Description:

The data set contain one year sales data from 1 April 2018 – 31 March 2019 which contains order details of customer, quantity of product purchased by the customer and profit details, Sales target data which contains target details of each category,

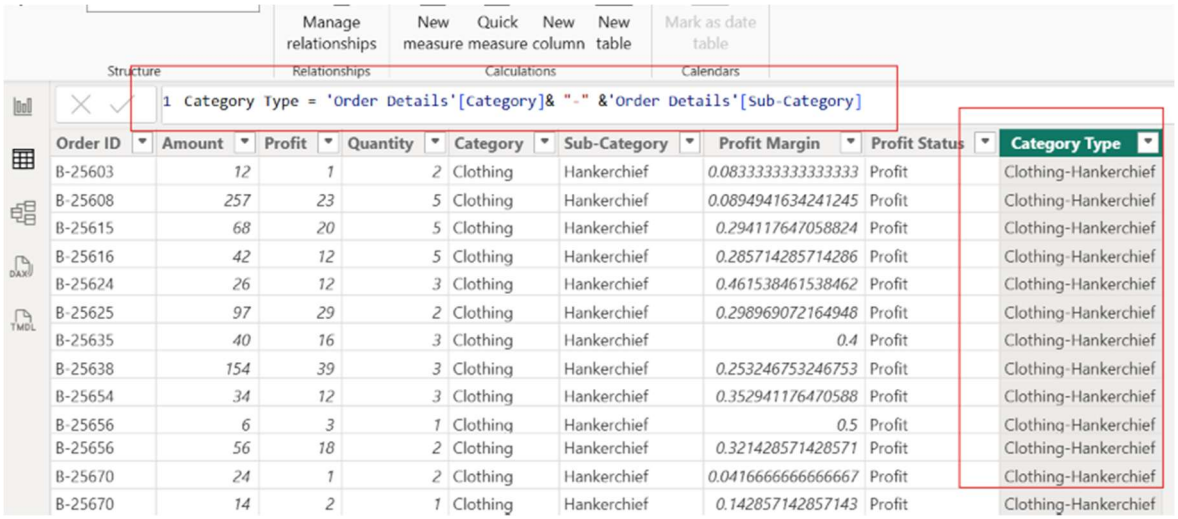
DAX Calculations:

Calculated Columns:

1. Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

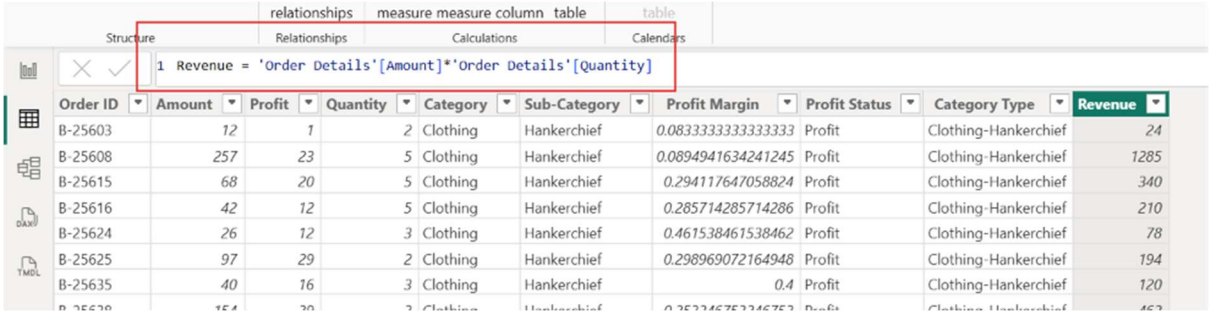


Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25851	336	71	3	Furniture	Bookcases	0.211309523809524	Profit
B-25852	869	67	4	Furniture	Tables	0.0771001150747986	Profit
B-25853	2093	721	5	Furniture	Chairs	0.344481605351171	Profit
B-25853	95	11	4	Furniture	Furnishings	0.115789473684211	Profit
B-25854	342	154	7	Furniture	Furnishings	0.450292397660819	Profit
B-25855	197	73	1	Furniture	Bookcases	0.370558375634518	Profit



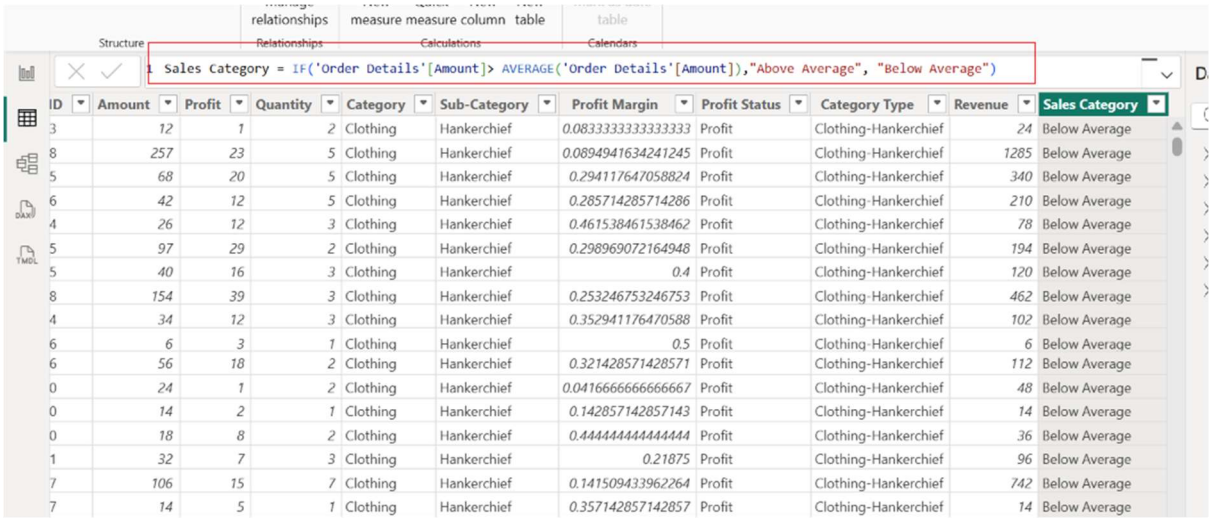
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing-Hankerchief
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing-Hankerchief
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing-Hankerchief
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing-Hankerchief
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing-Hankerchief
B-25625	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing-Hankerchief
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief
B-25638	154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing-Hankerchief
B-25654	34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing-Hankerchief
B-25656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing-Hankerchief
B-25656	56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing-Hankerchief
B-25670	24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit	Clothing-Hankerchief
B-25670	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing-Hankerchief

2. Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing-Hankerchief	24
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing-Hankerchief	1285
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing-Hankerchief	340
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing-Hankerchief	210
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing-Hankerchief	78
B-25625	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing-Hankerchief	194
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief	120

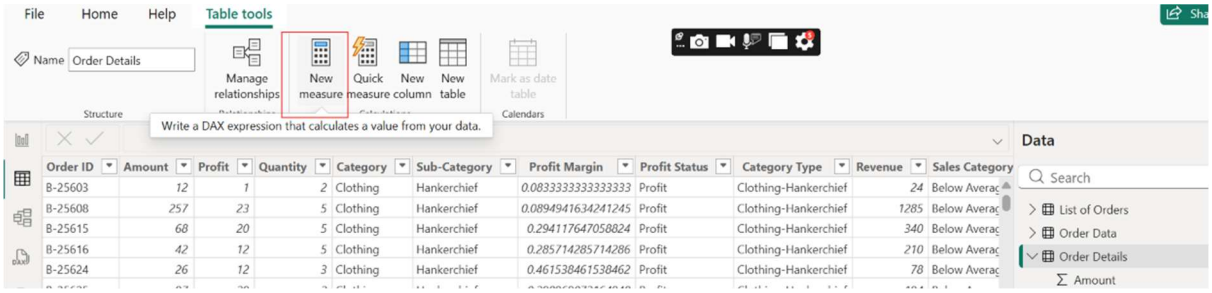
3. Create a Calculated Column to Categorize Sales: Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.



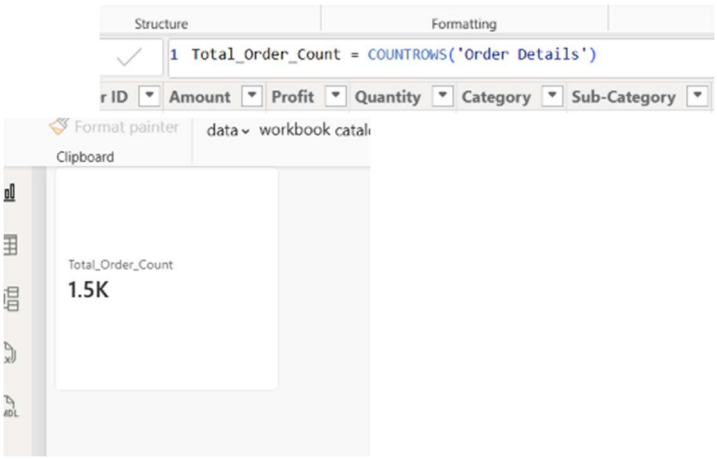
ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Sales Category
3	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing-Hankerchief	24	Below Average
8	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing-Hankerchief	1285	Below Average
5	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing-Hankerchief	340	Below Average
6	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing-Hankerchief	210	Below Average
4	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing-Hankerchief	78	Below Average
5	97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing-Hankerchief	194	Below Average
5	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief	120	Below Average
8	154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing-Hankerchief	462	Below Average
4	34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing-Hankerchief	102	Below Average
6	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing-Hankerchief	6	Below Average
6	56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing-Hankerchief	112	Below Average
0	24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit	Clothing-Hankerchief	48	Below Average
0	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing-Hankerchief	14	Below Average
0	18	8	2	Clothing	Hankerchief	0.444444444444444	Profit	Clothing-Hankerchief	36	Below Average
1	32	7	3	Clothing	Hankerchief	0.21875	Profit	Clothing-Hankerchief	96	Below Average
7	106	15	7	Clothing	Hankerchief	0.141509433962264	Profit	Clothing-Hankerchief	742	Below Average
7	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit	Clothing-Hankerchief	14	Below Average

Calculated Measures:

1. Order Count- Create a measure that calculates the total number of orders in the Order Details table.

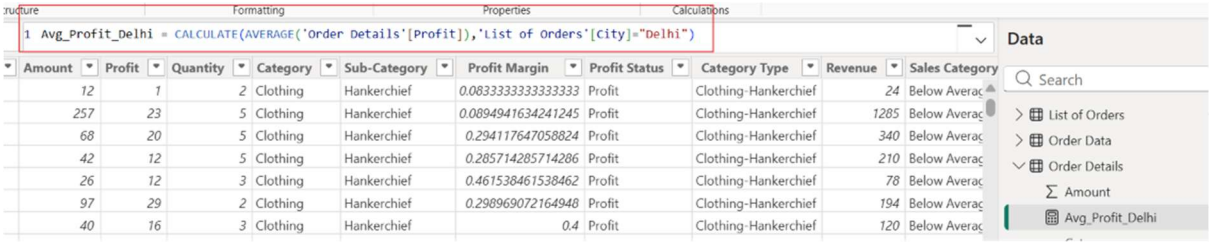


Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Sales Category
B-25603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing-Hankerchief	24	Below Averac
B-25608	257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing-Hankerchief	1285	Below Averac
B-25615	68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing-Hankerchief	340	Below Averac
B-25616	42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing-Hankerchief	210	Below Averac
B-25624	26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing-Hankerchief	78	Below Averac



2. Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi.

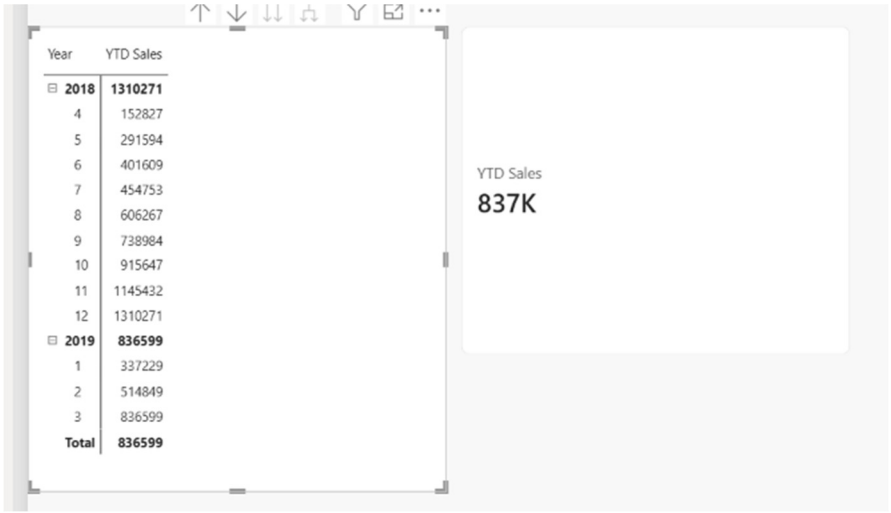




Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Sales Category
12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing-Hankerchief	24	Below Averag
257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing-Hankerchief	1285	Below Averag
68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing-Hankerchief	340	Below Averag
42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing-Hankerchief	210	Below Averag
26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing-Hankerchief	78	Below Averag
97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing-Hankerchief	194	Below Averag
40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief	120	Below Averag

3. Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.





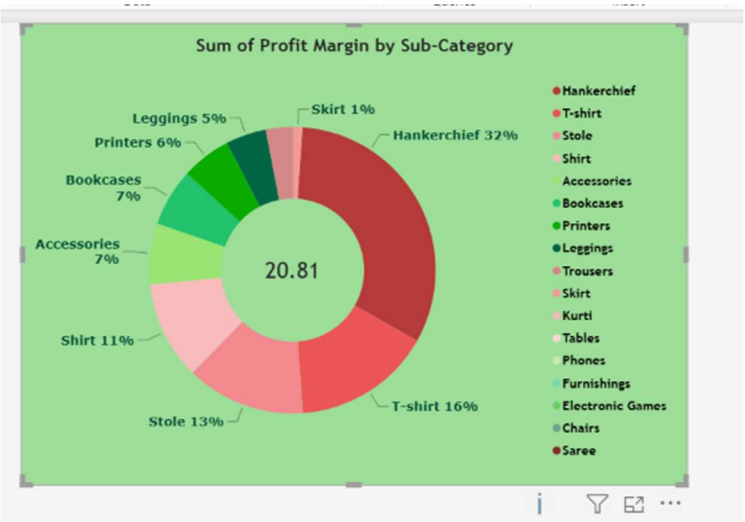
Year	YTD Sales
2018	1310271
4	152827
5	291594
6	401609
7	454753
8	606267
9	738984
10	915647
11	1145432
12	1310271
2019	836599
1	337229
2	514849
3	836599
Total	836599

Data Visualization:

1. Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.



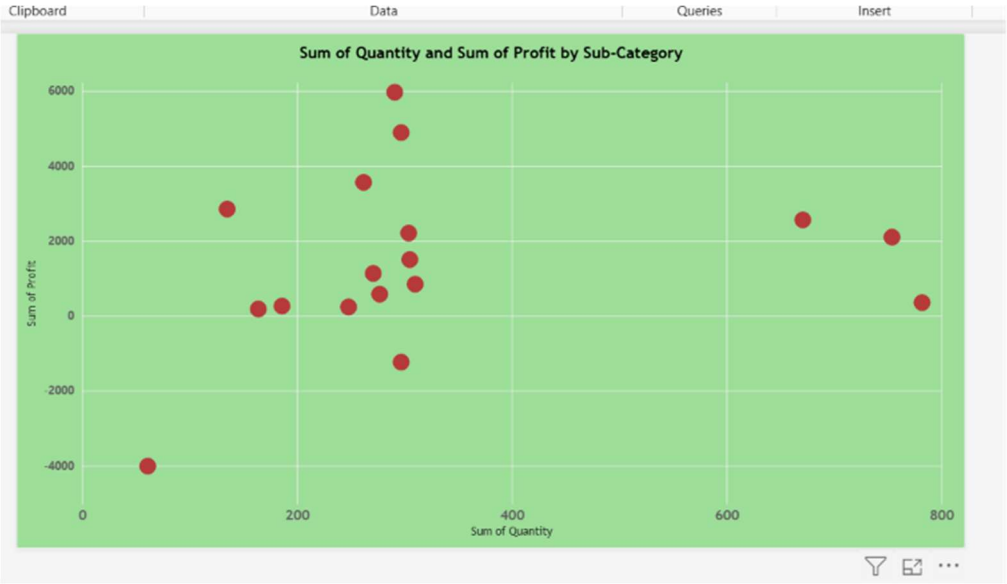
2. Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.



3. Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.



4. Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

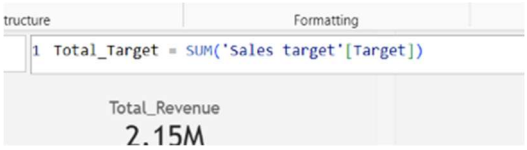


5. Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

   Dax measure Calculated below for Total Revenue :



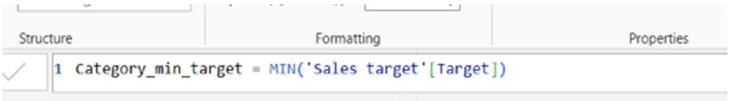
Dax measure Calculated for Total Target:



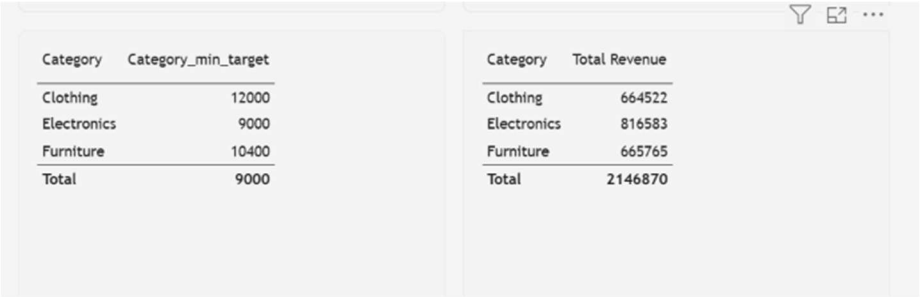
Comparison of Total Revenue and Total Target:



Dax Measure calculated for Min Target value:



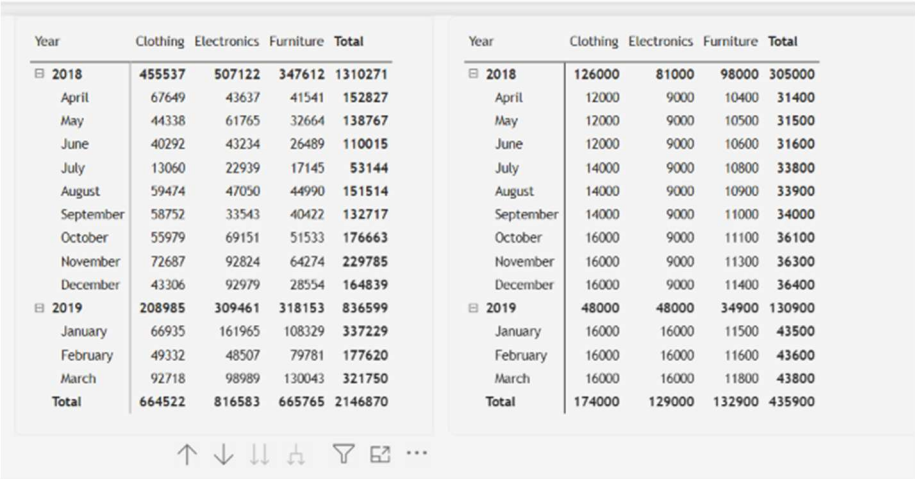
Added a table view from Visualization tab and added category and Mininum target value for each category:



Category	Category_min_target
Clothing	12000
Electronics	9000
Furniture	10400
Total	9000

Category	Total Revenue
Clothing	664522
Electronics	816583
Furniture	665765
Total	2146870

6. Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.



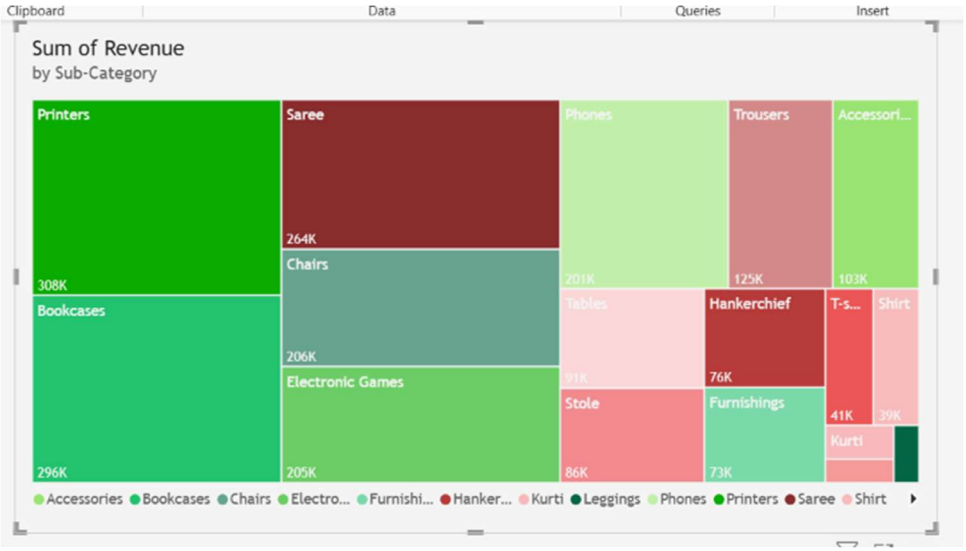
Year	Clothing	Electronics	Furniture	Total
2018	455537	507122	347612	1310271
April	67649	43637	41541	152827
May	44338	61765	32664	138767
June	40292	43234	26489	110015
July	13060	22939	17145	53144
August	59474	47050	44990	151514
September	58752	33543	40422	132717
October	55979	69151	51533	176663
November	72687	92824	64274	229785
December	43306	92979	28554	164839
2019	208985	309461	318153	836599
January	66935	161965	108329	337229
February	49332	48507	79781	177620
March	92718	98989	130043	321750
Total	664522	816583	665765	2146870

Year	Clothing	Electronics	Furniture	Total
2018	126000	81000	98000	305000
April	12000	9000	10400	31400
May	12000	9000	10500	31500
June	12000	9000	10600	31600
July	14000	9000	10800	33800
August	14000	9000	10900	33900
September	14000	9000	11000	34000
October	16000	9000	11100	36100
November	16000	9000	11300	36300
December	16000	9000	11400	36400
2019	48000	48000	34900	130900
January	16000	16000	11500	43500
February	16000	16000	11600	43600
March	16000	16000	11800	43800
Total	174000	129000	132900	435900

7. Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



8. Sales Distribution by Sub-Category: Represent the sales distribution across different sub-categories using a treemap.



9. Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.

